**SAMUEL B. SONNEGA**

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ssonnega@umassd.edu | **August, 2026**

## EDUCATION

-University of Massachusetts Dartmouth, PhD, Integrative Biology, May, 2026
-Ohio Wesleyan University, Honors Program, B.A. Zoology May, 2014

## RESEARCH AND WORK EXPERIENCE

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| Sep 2021- May 2026 | **PhD Researcher** in Michael Sheriff's lab at the University of Massachusetts Dartmouth. Designed and executed experiments to test the effects of predation risk on the gut microbiome of wild mice. Conducted fieldwork with white footed mice involving playback experiments, behavioral assays, and live trapping. Lab work included radioimmunoassay, microbial genetic DNA isolation, PCR, and library prep for Illumina sequencing. Performed extensive statistical analysis for several lab projects in addition to PhD work. |
| Sep 2016-April 2017 | **Intern** with The Nature Conservancy of Massachusetts. Organized and worked with abundance and distribution data collected by both TNC and Mass Fish and Wildlife, in order to formulate conservation plans for multiple species in Western Massachusetts. |
| May-Sep 2016 | **Research Assistant** to Dr. Sarah Partan, Assistant Professor, Hampshire College. Assisted in research exploring multimodal communication across multiple taxa and in writing and editing a review paper on that subject. |
| February-May 2016 | **Lead Biological Technician.** Kluane Red Squirrel Project |
| August-October 2015 | **Research Assistant** to PhD student Erin Siracusa, University of Guelph. Work I conducted involved territory removals, audio recording of vocalizations, and territorial vocalization playback experiments. |
| February-May 2015 | **Research Assistant** to Dr. Ben Dantzer, Assistant Professor of Psychology, University of Michigan. Conducted fieldwork investigating the maternal effects of the stress hormone cortisol on the American red squirrel, *Tamiasciurus hudsonicus.* Data collected included blood and fecal cortisol samples, pup growth rates, and focal behavioral observations. |
| July-October 2014 | **Biological Technician**. Kluane Red Squirrel Project. Work included live trapping, ear tagging, radio telemetry, and DNA sample collection. |
| 2013-2014 | **Research Assistant** to Tamara Panhuis, Assistant Professor of Zoology, Ohio Wesleyan University. Developed research on the relationship of female mate preference to MHC allelic diversity in the Sailfin Molly, *Poecilia latipinna.* Lab work I conducted included gel electrophoresis, DNA isolation, Nano-drop DNA analysis, PCR |

preparation, 454 platform pyrosequencing, fish dissection, scanning electron microscopy and preparation using glutaraldehyde fixation and $CO_2$ critical point drying.

## TEACHING EXPERIENCE

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| 2025-2026 | **Lecturer. (course design and lecture)** UMass Dartmouth.<br>-Advanced Ecology |
| 2021-2026 | **Teaching Assistant.** UMass Dartmouth. Courses TAed:<br>-Anatomy and Physiology<br>-Introductory Biology |
| 2017-2020 | **Teaching Assistant.** Trent University. Courses TAed:<br>- Introductory Ecology<br>- Behavioural Ecology<br>- Fisheries Ecology and Management |
| 2018-19 | **Guest Lectures.** Trent University.<br>-Behavioral ecology: predator prey dynamics<br>-Behavioral ecology: ecological physiology |
| 2019-2020 | **Course Design.** Trent University, Biodiversity and Ecological Monitoring Program.<br>-Applied Wildlife Endocrinology. |
| 2020 | **Workshops**<br>-Machine learning in R: use of support vector machines in ecology.<br>-Bayesian models in R |

## PUBLICATIONS

Nogueira, H., **Sonnega, S.**, DiNuzzo, E., Donelan, S., Sheriff, M.J. The influence of predator lethality on non-consumptive predation risk effects. *Ecology, in press.*

**Sonnega, S.** and Sheriff, M.J., 2024. Harnessing the gut microbiome: a potential biomarker for wild animal welfare. *Frontiers in Veterinary Science*, *11*, p.1474028.

Miller, H.A., Gobin, J., Boudreau, M.R., Horne, L.G., Scholl, L.E., Seguin, J.L., **Sonnega, S**., Krebs, C.J., Boonstra, R., Kenney, A.J. and Jung, T.S., 2024. Cyclic dynamics drive summer movement ecology of snowshoe hares (Lepus americanus). *Frontiers in Ecology and Evolution*, *12*, p.1419245.

Sheriff, M.J., Mancini, I., Aguiar, O.K., DiNuzzo, E.R., Maloney-Buckley, S., **Sonnega, S**. and Donelan, S.C., 2023. The impact of food availability on risk-induced trait responses in prey. *Behavioral Ecology*, *34*(6), pp.1036-1042.

Aguiar, O., **Sonnega, S.**, DiNuzzo, E.R. and Sheriff, M.J., 2023. Playing it safe; risk-induced trait responses increase survival in the face of predation. *Journal of Animal Ecology*, *92*(3), pp.690-697.

Gobin, J., Hossie, T.J., Derbyshire, R.E., **Sonnega, S**., Cambridge, T.W., Scholl, L., Kloch, N.D., Scully, A., Thalen, K., Smith, G. and Scott, C., 2022. Functional responses shape node and network level properties of a simplified boreal food web. *Frontiers in Ecology and Evolution*, *10*, p.898805.

Boudreau, M.R., Seguin, J.L., Lavergne, S.G., **Sonnega, S.**, Scholl, L., Kenney, A.J. and Krebs, C.J., 2020. Please come again: attractive bait augments recapture rates of capture-naïve snowshoe hares. *Wildlife Research*, *47*(3), pp.244-248.

***Manuscripts in progress***

**Sonnega, S.**, Bucci, V., Ganci, C., Silby, M., Sheriff, M.J. Perceived predation risk alters gut microbiome dynamics in wild captive *Peromyscus leucopus. In prep for Nature Ecology & Evolution*

**Sonnega, S**., Ganci, C., Zeamer, A., Mortzfeld, B., Bhattarai, S.K., Bucci, V., Sheriff, M.J. Fear runs wild: predation risk and gut microbial dynamics in free-living white-footed mice. *In prep for Ecology Letters.*

**Sonnega, S.** Boudreau, M.R., Seguin J.L., Boonstra, R., Palme, R., and Murray, D.L. Understanding trait variability in cyclic populations: do all metrics sing the same song? *In prep for Proceedings B*

## PRESENTATIONS

The influence of predator cues on gut microbial communities in white-footed mice: the perils and pitfalls of seeking a silver bullet welfare metric. NYU Wild Animal Welfare Summit, June 2026.

Predator cues to gut clues: Predation risk effects on the gut microbiome depend on captivity context in *Peromyscus leucopus*. Presentation and discussion leader at GRC predator-prey interactions, February 2026.

Incorporating citizen science (iNaturalist) into undergraduate ecology teaching. CUBICS Institute Annual Conference, UMass Dartmouth, December 2025.

From feces to function: the microbiome's role in assessing wild animal welfare. Presentation at the Annual Biology Conference, UMass Dartmouth, April 2025.

Going with the gut: ecology of the gut microbiome. Invited talk, University of New Hampshire biology department, March 2024.

How does predation risk affect the gut microbiome of wild white-footed mice (*Peromyscus leucopus*)? Poster presentation at GRC predator-prey interactions, February 2024.

Utilizing the gut microbiome as a welfare biomarker. Methods Workshop, Wild Animal Initiative, November 2023.

Disentangling the fitness payoffs and costs of risk induced trait responses. Presentation at ESA/CSEE Conference, August 2022.

Understanding trait variability in cyclic populations of snowshoe hare. Presentation at The Wildlife Society conference, March 2021.

Land use and conservation in Karukinka, Chile. Presentation to OWU faculty, students, and members of the Theory-to-Practice Board, April 2014.

Technology use in Kosrae, Micronesia. Paper presented at the Northeast Ohio Undergraduate Sociology Symposium, February 2014.

The People of Kosrae, Federated States of Micronesia. Presentation to OWU faculty, students, and members of the Theory-to-Practice Board, October 2014.

Mate Preference associated with MHC alleles in the sailfin molly, *Poecilia latipinna.* Poster presented at the Patricia Belt Conrades Summer Science Research Symposium, September 2013.

Evolutionary Foundations for Medicine and Public Health, Focus on Cancer and Infection, Mount Desert Island Biological Laboratories, Maine, Summer Course, August 2012.

## AWARDS AND GRANTS

**UMass Graduate Student Travel Grant.** $500. Jan 2026.

**Developing a consensus profile of wild animal welfare: integrating non-invasive monitoring of the gut microbiome with stress physiology and behavior.** $29,130. Wild Animal Initiative. June 2023.

**CUBICS Institute Grant.** $1200. Engaging students in the collection of biodiversity data through the use of citizen science tools.

**UMass Dartmouth Biology Department Grant.** $500. Sept. 2022.

**Green Bishop Award 2012-2013** for actions in campus sustainability efforts, e.g., applied for and secured funding through the Ohio Wesleyan Theory-to-Practice Grant Program for solar panels for the Environmental Studies House.

**A Unique Model for Land Conservation and Preservation in Karukinka, Chile**. $22,000 grant awarded competitively through the Ohio Wesleyan Theory-to-Practice Grant Program to **Sam Sonnega**, Melissa Guziak, and Owen Kelling (students), and William Hayes (staff). Travelled to Tierra Del Fuego, Chile for nine days in January 2014 to study the conservation biology and land-use practices of the newly formed Karukinka National Park.

**Ethnographic Research on Kosrae Island, Federated States of Micronesia.** $10,000 grant awarded competitively through the Ohio Wesleyan Theory-to-Practice Grant Program to James Peoples (faculty), **Sam Sonnega**, Erika Nininger, and Karena Briggs. Travelled to Kosrae, Micronesia to study technology use among the native people.

## SERVICE AND OUTREACH

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| 2024-present | Executive, UMass Dartmouth Graduate Student Union. Led organizing meetings and communications during mediation and contract negotiations. |
| 2019 | VP Communications for the Trent Graduate Students' Association |
| 2019 | Served on Senate Teaching Awards Committee |
| 2019-present | Member of Skype a Scientist |
| 2018 | Participated in Three Minute Thesis (3MT) competition (https://www.youtube.com/watch?v=ZThSgUNTRFg&t=58s) |
| 2017-18 | Organized science communication initiative in Haines Junction, Yukon. Presented research for school groups and members of the public. |
| 2018 | Served as Science Co-Chair for the Northern Studies Colloquium |
| 2012-2014 | Led weekly discussions in the community on local sustainability issues |

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| | (e.g., the meaning of wilderness, the role of science in conservation). |
| 2012-2014 | Obtained approval to initiate and run campus research apiary; took class at Ohio State University apiculture laboratory Summer 2013. |

## PROFESSIONAL AFFILIATIONS

Graduate Student Member of Ecological Society of America
Graduate Student Member of American Society of Mammalogists
Student Member of Arctic Institute of North America
Graduate Student Member of The Wildlife Society

## PEER REVIEW

Ecological and Evolutionary Physiology, reviewer.
Current Microbiology, reviewer.
Molecular Ecology, reviewer.
Oecologia, reviewer

## TECHNICAL SKILLS

*Field*- small mammal live-trapping and handling: morphometrics, ear-tagging, collaring, reproductive assessment. Working long hours in remote locations.
*Lab*- RIA/EIA analysis of hormonal metabolites, DNA extraction, library prep and sequencing (Illumina platform), bioinformatic pipelines for microbial genome assembly.
*Data science*- R: tidyverse, brms, glmmTMB, structural equation modeling, etc.
Python: OpenCV, DeepLabCut, automation and data tidying.
*Reproducible science*- Shiny app design, use of R Markdown and Quarto, Git version control.